



Optimal policy for inventory systems with capacity commitment and fixed ordering costs

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ARTICLE INFO

Article history:

Received 29 November 2018

Received in revised form 14 October 2019

Accepted 14 October 2019

Available online 23 October 2019

Keywords:

Inventory/production model

K -convexity

(s, S) policy

Capacity commitment

ABSTRACT

This paper investigates a single-product, periodic-review, non-stationary inventory system with total maximum capacity commitment and fixed ordering costs over a finite planning horizon. We characterize the optimal ordering policy as a state-dependent (s, S) policy whose parameters only depend on the sum of the net inventory and the remaining capacity. We show that such policy can degenerate into two simple policies in two special cases respectively. We also derive bounds on parameters of the optimal ordering policy.

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1. Introduction

Product suppliers are often constrained by how much they are able to produce in a given period of time. These constraints may be caused by conditions such as limited production capacity [7], ensuring product reach during promotion, and government regulations [11]. For instance, Shermag Inc., one of the largest Canadian furniture producer, signs annual contracts to procure logs from sawmills that operate in public forests. Those sawmills typically face annual quotas imposed by governments on the amount they can cut for each type of logs. The presence of annual quotas not only limits the logs supply from the sawmills, but also complicates the production planning of Shermag Inc. [15]. Similar examples also appear in other industries, such as the annual fishing allowance in some countries with overusing issues [11], and water usage allowance in areas with limited water resources [6], and carbon emission cap imposed on polluting companies [12]. As a result, these product suppliers often have to limit the purchase quantities from their buyers, a practice widely adopted by sellers such as those on Amazon and ebay, etc. We refer this as *total maximum capacity commitment* agreement, where the cumulative order quantities placed by the buyer over a given period of time cannot exceed a certain quantity.

Our work is also motivated by the growing literature on total maximum commitment in inventory literature. Though studying the total minimum commitment contract, [1] first points out the need to investigate the impact of the upper (maximum) total commitment on inventory management. [22] shows that the myopic base-stock policy is optimal to inventory systems with total maximum commitment, i.e., ordering up to the minimum between a base-stock level and an available inventory position that equals the on-hand inventory plus the remaining capacity. [23] further extends the results in [22] to systems with batch orders. [3] investigate a production–inventory system where the firm is subject to an allowance over the planning horizon, in addition to the fixed production capacity constraint in each period. A common assumption shared by this stream of literature is that there are only variable ordering costs but *no* fixed ordering costs. However, it is well-known that fixed ordering costs are common in production/inventory systems and can significantly alter the optimal ordering policy (even without capacity constraints) [e.g., 16,20]. It is unclear whether the myopic policy can remain optimal in inventory systems with capacity commitment in the presence of fixed ordering costs. This paper aims to fill this gap by considering fixed ordering costs in the inventory system with total maximum commitment, and characterizing its optimal capacity commitment level and ordering policies.

In this paper, we consider a single-product, periodic-review, non-stationary inventory system with fixed ordering costs and capacity commitment, where the buyer's total order quantity during the whole (finite) selling season cannot exceed the committed level that is determined prior to the selling season. In each selling

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period, the buyer orders from the supplier to satisfy the stochastic demand subject to the remaining capacity. The buyer's objective is to minimize the total expected discounted cost throughout the selling season, including the cost of purchasing capacity, and the ordering and inventory-related costs. Using the notion of K -convexity, we show that the optimal ordering policy is a state-dependent (s, S) policy that only depends on the sum of the net inventory and the remaining capacity in each period. We further show that the state-dependent (s, S) policy can be simplified as the myopic base-stock policy when there are no fixed ordering costs and as the classical stationary (s, S) policy when there is leftover capacity, respectively. In this sense, we generalize the myopic base-stock policy and the classical (s, S) policy. In particular, we characterize the effect of the remaining capacity level on the corresponding optimal ordering quantity in each period by employing $L^\square$ -convexity. Finally, we derive bounds on the parameters of the optimal policy, which are useful for computational purposes.

Our work directly contributes to the increasing literature on total maximum commitment [e.g., 3,22,23] as aforementioned. Other forms of capacity commitment includes flexible commitment, where the buyer is allowed to adjust the commitments over time [e.g., 2,4,19], and total minimum commitment, where the buyer's total order quantities cannot be less than the committed level [e.g., 8,9,24]. Our work also contributes to the inventory literature on characterizing the structure of the optimal policy for inventory systems with fixed ordering costs. The notion of K -convexity is first introduced by [16] to show the optimality of the (s, S) policy for one single-product inventory system with fixed costs but without commitment. Thereafter, K -convexity has played a critical role in deriving the structure of the optimal policy for inventory systems with fixed ordering costs under various settings [see e.g., 10,17,18, and references therein]. In addition, [20] provides an alternative proof for the (s, S) policy based on the quasi-convexity and [21] gives a complete computational approach to find the bounds and optimal values of the two parameters s, S . To the best of our knowledge, our work is the first attempt to characterize the ordering policy with total maximum commitment and fixed ordering costs.

The organization of the remainder is as follows. Section 2 presents the model formulation. Section 3 analyzes the structure of the optimal policy. Section 4 discusses two special cases of the optimal policy. Section 5 derives bounds on parameters of the optimal policy and Section 6 presents the numerical studies. We conclude in Section 7 by discussing possible extensions.

2. Model formulation

Consider the buyer managing a single-product periodic-review inventory system over a finite selling season consisting of T periods. The buyer can order from the supplier to satisfy the nonnegative random demand D_t in each period t for $t = 1, \dots, T$. Each order incurs the fixed ordering cost K and the variable cost with the unit cost c . We assume that demands in different periods are independent (but not necessarily identically) distributed and the deliver is instant once the order is placed, i.e., there is no leadtime.

Prior to the selling season, the buyer signs a long-term contract with the supplier to make a total maximum capacity commitment with level C_0 . That is, the buyer commits that the total ordering quantity over the selling season cannot exceed the committed capacity C_0 . Let $P(C_0)$ denote the cost for purchasing C_0 units of the capacity.

Once the commitment is confirmed, the sequence of events in each period $t = 1, \dots, T$ is as follows:

- (1) At the beginning of period t , the buyer first reviews the initial inventory level x_t and the remaining capacity C_t (note that $C_1 = C_0$), then orders q_t from the supplier. The order q_t cannot exceed the remaining capacity nor be negative, i.e., $0 \leq q_t \leq C_t$. Then buyer incurs the ordering cost

$$c(q_t) = K\delta(q_t) + cq_t,$$

where $\delta(q) = 1$ if $q > 0$ and $\delta(q) = 0$ otherwise. The order placed arrives immediately and raises the inventory level up to $y_t = x_t + q_t$. We refer to y_t as the post-order inventory level in period t .

- (2) The demand D_t is realized and fulfilled as much as possible. The unsatisfied demand $[D_t - y_t]^+$ is backlogged with unit shortage cost b , where $[x]^+ = \max\{x, 0\}$. The excess inventory $[y_t - D_t]^+$ is also carried over to the next period with the unit holding cost h . We assume that $b > c$, that is, the buyer has no incentive to backlog inventory intentionally.
- (3) At the end of period t , all costs incurred in period t are calculated.

The objective of the buyer is to minimize the total expected discounted cost throughout the selling season, including the cost of purchasing capacity prior to the selling season and the ordering, shortage and holding costs incurred during the selling season. Let $\alpha \in (0, 1]$ denote the discount factor.

We shall formulate the buyer's problem as a dynamic program. Define $z_t = x_t + C_t$ as the aggregate inventory position at period t and define the vector (x_t, z_t) as the state of the system. The system dynamics from period t to period $t + 1$ are given by

$$x_{t+1} = x_t + q_t - D_t = y_t - D_t, \quad z_{t+1} = z_t - D_t.$$

Let $v_t(x_t, z_t)$ denote the minimum expected total discounted cost from period t and onward given the state (x_t, z_t) in period t and let $T + 1$ denote the end of the period T , or the terminal period. Then the dynamic recursion for $v_t(x_t, z_t)$ can be formulated as, for $t = 1, \dots, T$,

$$v_t(x_t, z_t) = \min_{x_t \leq y_t \leq z_t} \{K\delta(y_t - x_t) + c(y_t - x_t) + G_t(y_t) + \alpha \mathbb{E}[v_{t+1}(y_t - D_t, z_t - D_t)]\}, \quad (1)$$

where

$$G_t(y_t) = \mathbb{E}\{h[y_t - D_t]^+ + b[D_t - y_t]^+\},$$

denotes the expected holding and shortage cost in period t , and the terminal cost is given by

$$v_{T+1}(x_{T+1}, z_{T+1}) = \tau_c(z_{T+1} - x_{T+1}) - \tau_l[x_{T+1}]^+ + \tau_b[-x_{T+1}]^+,$$

where τ_c denotes the unit cost of the unused capacity, τ_l denotes the unit salvage value of leftovers, and τ_b denotes the unit shortage cost of unmet demand at the end of period T . It is reasonable to assume that $\tau_l < c < \tau_b$. We also assume $\tau_c \geq c - \tau_l$ so that the buyer has no incentive to reserve excessive capacity.

By solving the dynamic programming in (1), the buyer can determine the optimal ordering policy in periods $1, \dots, T$, and calculate $v_1(x_1, z_1)$, i.e., the minimum sum of the expected ordering, shortage and holding cost given the capacity commitment $C_1 = C_0$. Then, the buyer can choose the optimal capacity C_0^* purchased prior to the selling season by solving the following problem

$$\min_{C_0 \geq 0} \{P(C_0) + v_1(x_1, z_1)\}. \quad (2)$$

For example, suppose $P(C_0) = pC_0$, where p denotes the unit price of purchasing capacity and assume $p > c$ to avoid exaggerating capacity commitment intentionally. The buyer's optimal capacity commitment is given by

$$C_0^* = \arg \min_{C_0 \geq 0} \{pC_0 + v_1(x_1, z_1 + C_0)\}.$$

Therefore, in what follows, we shall focus on the buyer's optimal ordering policy.

Note that the ordering decision in each period affects not only the net inventory level but also the remaining capacity in the next period. This significantly complicates the problem as $C_1, \dots, C_T$ are now endogenized. In the presence of fixed ordering costs, it is well-known that for systems without capacity constraints, the so-called (s, S) policy is optimal, i.e., there exist two scalars s_t and S_t for each period t , such that it is optimal to order up to S_t if the inventory level is below s_t and to order nothing otherwise in that period. However, it is still unclear what kind of policy is optimal for such systems when total maximum capacity commitment is also present. One interesting research question is whether the optimal policy is similar to the classical (s, S) policy. Also note that K -convexity is central to show the optimality of the (s, S) policy, see [16], however, this property generally does not preserve under the minimization operator. Another question is whether K -convexity can be preserved or how to preserve it in inventory systems with fixed ordering costs and capacity commitment.

3. The main results

In this section, we characterize the optimal ordering policy by utilizing the notion of K -convexity. To facilitate the subsequent analysis, we first review two alternative definitions and several preliminary results of K -convexity.

Definition 1. A real-valued function $g : \mathbb{R} \rightarrow \mathbb{R}$ is said to be K -convex ($K \geq 0$) if for any $z \geq 0$, $b > 0$ and any y , the following inequality holds:

$$K + g(z + y) \geq g(y) + \frac{z}{b}[g(y) - g(y - b)]. \quad (3)$$

Definition 2 ([17]). A real-valued function $g : \mathbb{R} \rightarrow \mathbb{C}$ is said to be K -convex if (3) holds whenever $z + y, y, y - b \in \mathbb{C}$, where $\mathbb{C}$ is a convex subset of $\mathbb{R}$.

Lemma 1. (1) A real-valued convex function is also 0-convex, and hence K -convex for $K \geq 0$. A K_1 -convex function is also K_2 -convex for $K_2 \geq K_1$.

(2) If $g_1(x)$ is K_1 -convex and $g_2(x)$ is K_2 -convex, then for any $\alpha_1, \alpha_2 \geq 0$, $\alpha_1 g_1(x) + \alpha_2 g_2(x)$ is $(\alpha_1 K_1 + \alpha_2 K_2)$ -convex.

(3) If $g(x)$ is K -convex and D is a random variable, then $\mathbb{E}[g(x - D)]$ is also K -convex, provided $\mathbb{E}[|g(x - D)|] < \infty$ holds for all x .

(4) If $g(x)$ is a continuous K -convex function with $g(x) \rightarrow \infty$ as $x \rightarrow \infty$, then there exist scalars s and S with $s \leq S$ such that

- (i) $g(S) \leq g(x)$ for all x ;
- (ii) $g(S) + K = g(s) < g(x)$ for all $x < s$;
- (iii) $g(x)$ is a nonincreasing function on $(-\infty, s)$; and
- (iv) $g(x) \leq g(y) + K$ for all x, y with $s \leq x \leq y$.

Lemma 2 ([17]). Let $g(x)$ be a continuous K -convex function with $g(x) \rightarrow \infty$ as $x \rightarrow \infty$. Let A and B be two extended real numbers for a closed interval $[A, B]$ which becomes open at A (or B) whenever $A = -\infty$ (or $B = \infty$). Define two extended real numbers S and s with $S \geq s$, which are given by

$$S = \min \arg \min_{A \leq x \leq B} g(x),$$

$$s = \min\{x | g(x) \leq g(S) + K, A \leq x \leq S\}.$$

Thus, the following properties hold:

- (1) $g(x) \leq g(y) + K$ for all x, y with $s \leq x \leq y \leq B$;
- (2) let $h(x) = \min_{y \geq x, A \leq y \leq B} \{K\delta(y - x) + g(y)\}$ on $(-\infty, B]$, then

$$h(x) = \begin{cases} K + g(S) & \text{for } x < s, \\ g(x) & \text{for } s \leq x \leq B, \end{cases}$$

and $h(x)$ is K -convex on $(-\infty, B]$; and

- (3) if $s > A$, then $g(x)$ is nonincreasing on $[A, s]$.

For the above two definitions of K -convexity, Definition 1 is canonical in the literature whereas Definition 2 is introduced by [17] to facilitate the application of K -convexity on convex sets, e.g., Lemma 2 extends property (4) of Lemma 1 to real intervals. Lemma 1 presents some well-known properties of K -convexity while Lemma 2 is critical to our results as it directly establishes the optimality of (s, S) -type policy (see Remark 4.4 of [17] for more detailed discussions). With these properties on hand, we can readily characterize the structure of the optimal ordering policy for the problem in (1).

To facilitate the analysis, we first define $f_t(x_t, z_t) = v_t(x_t, z_t) + \alpha x_t$ and reformulate the dynamic recursion in (1) as follows, for $t = 1, \dots, T$,

$$f_t(x_t, z_t) = \min_{x_t \leq y_t \leq z_t} \{G_t(y_t) + (1 - \alpha)cy_t + \alpha c\mu_t + K\delta(y_t - x_t) + \alpha \mathbb{E}[f_{t+1}(y_t - D_t, z_t - D_t)]\}, \quad (4)$$

where the terminal cost is given by

$$f_{T+1}(x_{T+1}, z_{T+1}) = \tau_c z_{T+1} + (c - \tau_c)x_{T+1} - \tau_l[x_{T+1}]^+ + \tau_b[-x_{T+1}]^+.$$

Similar to Lemma 2, define parameters $S_t(z_t)$ and $s_t(z_t)$ with $S_t(z_t) \geq s_t(z_t)$, i.e.,

$$S_t(z_t) = \min \arg \min_{-\infty < y_t \leq z_t} U_t(y_t, z_t), \quad (5)$$

$$s_t(z_t) = \min\{y_t | U_t(y_t, z_t) \leq U_t(S_t(z_t), z_t) + K, -\infty < y_t \leq S_t(z_t)\}, \quad (6)$$

where

$$U_t(y_t, z_t) = G_t(y_t) + (1 - \alpha)cy_t + \alpha c\mu_t + \alpha \mathbb{E}[f_{t+1}(y_t - D_t, z_t - D_t)]. \quad (7)$$

That is, $S_t(z_t)$ is the smallest global minimizer of $U_t(y_t, z_t)$, $-\infty < y_t \leq z_t$, and $s_t(z_t)$ represents the minimum point whose cost is no greater than the cost of ordering up to $S_t(z_t)$.

The following theorem fully characterizes the structure of the optimal ordering policy for the problem in (4).

Theorem 3. (1) For $t = 1, \dots, T$, $f_t(x_t, z_t)$ is K -convex in x_t on $(-\infty, z_t]$.

(2) For $t = 1, \dots, T$, the optimal ordering policy in period t is a state-dependent (s, S) policy: given that x_t and z_t are the inventory level and the aggregate inventory position in period t , respectively, there exist parameters $s_t(z_t)$ and $S_t(z_t)$ with $s_t(z_t) \leq S_t(z_t) \leq z_t$ such that if $x_t < s_t(z_t)$, then it is optimal to order up to $S_t(z_t)$, otherwise it is optimal to order nothing.

Proof. We prove property (1) by induction. For $t = T + 1$, recall that

$$\begin{aligned} f_{T+1}(x_{T+1}, z_{T+1}) &= \tau_c z_{T+1} + (c - \tau_c)x_{T+1} - \tau_l[x_{T+1}]^+ + \tau_b[-x_{T+1}]^+ \\ &= \begin{cases} \tau_c z_{T+1} + (c - \tau_c - \tau_l)x_{T+1} & \text{for } x_{T+1} \geq 0; \\ \tau_c z_{T+1} + (c - \tau_c - \tau_b)x_{T+1} & \text{for } x_{T+1} < 0. \end{cases} \end{aligned}$$

It is clear that $f_{T+1}(x_{T+1}, z_{T+1})$ is convex and hence K -convex by property (1) of Lemma 1. Furthermore, based on properties (1–3) of Lemma 1, $U_T(y_T, z_T)$ is K -convex in y_T on $(-\infty, z_T]$ for any z_T . Therefore, it follows from property (2) of Lemma 2 that $f_T(x_T, z_T) = \min_{x_T \leq y_T \leq z_T} \{U_T(y_T, z_T) + K\delta(y_T - x_T)\}$ is K -convex on $(-\infty, z_T]$.

For $t = 1, \dots, T$, assuming that $f_{t+1}(x_{t+1}, z_{t+1})$ is K -convex in x_{t+1} on $(-\infty, z_{t+1}]$ for any $z_{t+1} \leq x_t + C_t$, then it suffices to show that $f_t(x_t, z_t)$ is also K -convex in x_t on $(-\infty, z_t]$ for any $z_t \leq x_{t-1} + C_{t-1}$. As $f_{t+1}(x_{t+1}, z_{t+1})$ is K -convex on $(-\infty, z_{t+1}]$

by the inductive assumption, it follows from properties (1–3) of [Lemma 1](#) that $U_t(y_t, z_t) = G_t(y_t) + (1 - \alpha)cy_t + \alpha c\mu_t + \alpha \mathbb{E}[f_{t+1}(y_t - D_t, z_t - D_t)]$ is K -convex in y_t on $(-\infty, z_t]$ for $z_t \leq x_{t-1} + C_{t-1}$. Then it follows from property (2) of [Lemma 2](#) that $f_t(x_t, z_t)$ is K -convex in x_t on $(-\infty, z_t]$. This completes the induction.

For property (2), given $S_t(z_t)$ and $s_t(z_t)$ in (5) and (6), respectively, it follows from [Lemma 2](#) that

$$f_t(x_t, z_t) = \begin{cases} K + U_t(S_t(z_t), z_t) & \text{for } x_t < s_t(z_t), \\ U_t(x_t, z_t) & \text{for } s_t(z_t) \leq x_t \leq z_t. \end{cases}$$

That is, a state-dependent (s, S) policy is optimal for the problem in (4). $\square$

It is worth noting that under the optimal state-dependent (s, S) policy, the ordering decision in each periods is similar to the classical (s, S) policy in [16]. That is, in period t , the buyer's ordering decision can be described by two parameters: $s_t(z_t)$ and $S_t(z_t)$, and the buyers orders up to $S_t(z_t)$ if the inventory level x_t is below $s_t(z_t)$. However, the state-dependent (s, S) policy is much more complex than the classical (s, S) policy as the two parameters $\{s_t(z_t), S_t(z_t)\}$ depend on the aggregate inventory position (i.e., the inventory level x_t plus the remaining capacity C_t). It implies that at the beginning of each period t the buyer needs not only to review the inventory level x_t but also to review the remaining capacity C_t , and makes ordering decision only relying on the aggregate information.

4. Two special cases

In this section, we discuss how the state-dependent (s, S) policy relates to the well-known policies in the literature. In particular, we show that the state-dependent (s, S) policy reduces to the myopic base-stock policy in [22] when there is no fixed-ordering cost, i.e., $K = 0$, and to the classical (s, S) policy in [16] when there is leftover capacity, i.e., capacity has never been used up, respectively. In this sense, the state-dependent (s, S) policy can be viewed as the generalization of these two policies.

4.1. The case without fixed ordering costs

In this case, we show that if there is no fixed ordering cost, i.e., $K = 0$, the state-dependent (s, S) policy can be simplified as the myopic base-stock policy suggested in [3,22,23].

With other settings remaining the same as in Section 3 except for $K = 0$, the dynamic programming in (4) can be reformulated as follows, for $t = 1, \dots, T$

$$\bar{f}_t(x_t, z_t) = \min_{x_t \leq y_t \leq z_t} \{G_t(y_t) + (1 - \alpha)cy_t + \alpha c\mu_t + \alpha \mathbb{E}[\bar{f}_{t+1}(y_t - D_t, z_t - D_t)]\},$$

with the terminal cost $\bar{f}_{T+1}(x_{T+1}, z_{T+1}) = \tau_c z_{T+1} + (c - \tau_c)x_{T+1} - \tau_l[x_{T+1}]^+ + \tau_b[-x_{T+1}]^+$. The following lemma is useful to establish the myopic base-stock policy, which is a special case of [13].

Lemma 4 (Karush, 1959 [13]). *If a real-valued function $g : \mathbb{R} \rightarrow \mathbb{R}$ is convex and attains its unconstrained minimum at $y_0 \in \mathbb{R}$, then, given $x \leq z$, $x \in \mathbb{R}$ and $z \in \mathbb{R} \cup \{+\infty\}$,*

$$\min_{x \leq y \leq z} g(y) = g_1(x) + g_2(z),$$

where $g_1(x) = \min_{x \leq y} g(y) = g(\max\{x, y_0\})$ is convex and nondecreasing in x , $g_2(z) = g(z) - \min_{x \leq y} g(y) = g(z) - g(\max\{z, y_0\})$ is convex and nonincreasing in z and $g_2(+\infty) = 0$ by convention.

[Lemma 4](#) states that the minimum of the convex function $g(y)$ on the convex set $[x, z]$ is additively convex in x, z and monotonic in x and z respectively. Applying [Lemma 4](#) to the buyer's dynamic programming problem above, we can show that the value function $\bar{f}_t(x_t, z_t)$ is also additively convex in x_t, z_t and monotonic in x_t and z_t respectively. It directly follows that the myopic base-stock policy is optimal. The following theorem fully characterizes the structure of the myopic base-stock policy.

Theorem 5. (1) For $t = 1, \dots, T$, $\bar{f}_t(x_t, z_t)$ is convex in x_t on $(-\infty, z_t]$.

(2) For $t = 1, \dots, T$, $\bar{f}_t(x_t, z_t) = f_{1,t}(x_t) + f_{2,t}(z_t)$, where $f_{1,t}(x_t)$ is convex and nondecreasing in x_t , and $f_{2,t}(z_t)$ is convex and nonincreasing in z_t .

(3) For $t = 1, \dots, T$, the optimal ordering policy in period t is a myopic base-stock policy: there exists a scalar S_t such that if $x_t < S_t$, then it is optimal to order up to $\min\{S_t, z_t\}$, otherwise it is optimal to order nothing.

Proof. We first show part (i) and (ii) by induction simultaneously.

For $t = T$, it is clear that $\bar{f}_{T+1}(x_{T+1}, z_{T+1}) = f_{T+1}(x_{T+1}, z_{T+1})$ is convex in x_{T+1} and decomposable in x_{T+1}, z_{T+1} . Then, it follows from [13] that $\bar{f}_T(x_T, z_T) = \min_{x_T \leq y_T \leq z_T} \{G_T(y_T) + (1 - \alpha)cy_T + \alpha c\mu_T + \alpha \mathbb{E}[\bar{f}_{T+1}(y_T - D_T, z_T - D_T)]\}$ is convex in x_t and decomposable in x_T, z_T .

For $t = 1, \dots, T - 1$, suppose that $\bar{f}_{t+1}(x_{t+1}, z_{t+1})$ is convex in x_{t+1} on $(-\infty, z_{t+1}]$ for any $z_{t+1} \leq x_t + C_t$, and decomposable in x_{t+1} and z_{t+1} , then it suffices to show that (i) and (ii) are also true for $\bar{f}_t(x_t, z_t)$. By the inductive assumption, we have $\bar{f}_{t+1}(x_{t+1}, z_{t+1}) = f_{1,t+1}(x_{t+1}) + f_{2,t+1}(z_{t+1})$, where $f_{1,t+1}(x_{t+1})$ is convex and nondecreasing in x_{t+1} , and $f_{2,t+1}(z_{t+1})$ is convex and nonincreasing in z_{t+1} . It follows that

$$\begin{aligned} \bar{f}_t(x_t, z_t) &= \min_{x_t \leq y_t \leq z_t} \{G_t(y_t) + (1 - \alpha)cy_t + \alpha c\mu_t + \alpha \mathbb{E}[\bar{f}_{t+1}(y_t - D_t, z_t - D_t)]\} \\ &= \min_{x_t \leq y_t \leq z_t} \{G_t(y_t) + (1 - \alpha)cy_t + \alpha c\mu_t + \alpha \mathbb{E}[f_{1,t+1}(y_t - D_t) + f_{2,t+1}(z_t - D_t)]\} \\ &= \min_{x_t \leq y_t \leq z_t} \{G_t(y_t) + (1 - \alpha)cy_t + \alpha \mathbb{E}[f_{1,t+1}(y_t - D_t)]\} + \alpha c\mu_t + \alpha \mathbb{E}[f_{2,t+1}(z_t - D_t)]. \end{aligned}$$

Since $G_t(y_t) + (1 - \alpha)cy_t + \alpha \mathbb{E}[f_{1,t+1}(y_t - D_t)]$ is convex, then by [Lemma 4](#), we have

$$\min_{x_t \leq y_t \leq z_t} \{G_t(y_t) + (1 - \alpha)cy_t + \alpha \mathbb{E}[f_{1,t+1}(y_t - D_t)]\} = g_{1,t}(x_t) + g_{2,t}(z_t),$$

where $g_{1,t}(x_t)$ is convex and nondecreasing in x_t , and $g_{2,t}(z_t)$ is convex and nonincreasing in z_t . Thus, we have

$$\bar{f}_t(x_t, z_t) = g_{1,t}(x_t) + g_{2,t}(z_t) + \alpha c\mu_t + \alpha \mathbb{E}[f_{2,t+1}(z_t - D_t)].$$

Let $f_{1,t}(x_t) = g_{1,t}(x_t)$ and $f_{2,t}(z_t) = g_{2,t}(z_t) + \alpha c\mu_t + \alpha \mathbb{E}[f_{2,t+1}(z_t - D_t)]$. Hence, $\bar{f}_t(x_t, z_t) = f_{1,t}(x_t) + f_{2,t}(z_t)$, where $f_{1,t}(x_t)$ is convex and nondecreasing in x_t , and $f_{2,t}(z_t)$ is convex and nonincreasing in z_t . This completes the induction.

We next show the optimality of the myopic base-stock policy. Denote $W_t(y_t) = G_t(y_t) + (1 - \alpha)cy_t + \alpha \mathbb{E}[f_{1,t+1}(y_t - D_t)]$. Let S_t be the global minimum of $W_t(y_t)$, i.e.,

$$S_t = \min_{-\infty < y_t \leq \infty} W_t(y_t).$$

Then, we have

$$\bar{f}_t(x_t, z_t) = \begin{cases} W_t(x_t) + \alpha c\mu_t + \alpha \mathbb{E}[f_{2,t+1}(z_t - D_t)] & \text{if } S_t \leq x_t, \\ W_t(\min\{S_t, z_t\}) + \alpha c\mu_t + \alpha \mathbb{E}[f_{2,t+1}(z_t - D_t)] & \text{otherwise.} \end{cases}$$

It follows that the myopic base-stock policy is optimal with the optimal order-up-to level $\min\{S_t, z_t\}$. $\square$

Note that the myopic base-stock policy is much simpler than the state-dependent (s, S) policy characterized in Theorem 3. To see this, notice that S_t does not depend on (x_t, z_t) . That is the optimal order-up-to level $\min\{S_t, z_t\}$ is not state-dependent but bounded by the aggregate inventory position z_t . The intuition is due to the additive convexity of the value function $\tilde{f}(x_t, z_t)$ in (x_t, z_t) , which allows us to specify S_t irrespective of z_t . Unfortunately, such property is unable to preserve in the presence of fixed ordering cost. It is also worth noting that [22] and [3] both show the optimality of the myopic base-stock policy by considering positive lead times and additional fixed capacity constraints in each periods to the total allowance on capacity, respectively. However, neither of them consider the fixed ordering cost.

4.2. The case with leftover capacity

In this section, we investigate how the remaining capacity affects the optimal ordering quantity in each period. In particular, we show that the optimal ordering policy can be simplified as the stationary (s, S) policy when there is leftover capacity, i.e., capacity has never been used up. It applies to scenarios that C_0^* is larger than the highest possible total demand over the entire planning horizon, which is possible if demand distribution is capped above.

To directly observe the impact of the remaining capacity, it is convenient to use (x_t, C_t) as the state variable. Let $v_t(x_t, C_t)$ denote the minimum expected cost from period t and onward, and $q_t^*(x_t, C_t)$ be the optimal order quantity in period t . We can rewrite the buyer's problem in (4) as follows, for $t = 1, \dots, T$,

$$v_t(x_t, C_t) = \min_{0 \leq q_t \leq C_t} \{G_t(x_t + q_t) + K\delta(q_t) + cq_t + \alpha \mathbb{E}[v_{t+1}(x_t + q_t - D_t, C_t - q_t)]\}, \quad (8)$$

with the terminal cost $v_{T+1}(x_{T+1}, C_{T+1}) = \tau_c C_{T+1} - \tau_l [x_{T+1}]^+ + \tau_b [-x_{T+1}]^+$.

The following lemma characterizes the $L^\natural$ -convexity of the value function $v_t(x_t, C_t)$ and presents some properties of the optimal order quantity $q_t^*(x_t, C_t)$ with respect to the remaining capacity C_t in each period t .

Lemma 6. For $t = 1, \dots, T$,

- (1) $v_t(x_t, C_t)$ is $L^\natural$ -convex in (x_t, C_t) ;
- (2) $q_t^*(x_t, C_t)$ is nondecreasing in C_t ;
- (3) $q_t^*(x_t, C_t + \xi) \leq q_t^*(x_t, C_t) + \xi$ for any $\xi > 0$.

Proof. We first show the $L^\natural$ -convexity of $v_t(x_t, C_t)$, then property (2) and (3) directly hold due to Lemma 3 of [25]. To follow the definition of $L^\natural$ -convexity and techniques introduced in [25], we first transform the states: define state variables as (x_t, B_t) and decision variable as ζ_t by letting $B_t = -C_t$ and $\zeta_t = -q_t$. Let $w_t(x_t, B_t) = v_t(x_t, B_t)$, then the dynamic programming in (8) becomes

$$w_t(x_t, B_t) = \min_{0 \leq -\zeta_t \leq -B_t} \{G_t(x_t - \zeta_t) - c\zeta_t + K\delta(-\zeta_t) + \beta \mathbb{E}[v_{t+1}(x_t - \zeta_t - D_t, B_t - \zeta_t)]\}$$

with $w_{T+1}(x_{T+1}, B_{T+1}) = -\tau_c B_{T+1} - \tau_l [x_{T+1}]^+ + \tau_b [-x_{T+1}]^+$. Then, it suffices to show $w_t(x_t, B_t)$ is $L^\natural$ -convex in (x_t, B_t) .

We show this by induction. It is clear that $w_{T+1}(x_{T+1}, B_{T+1})$ is $L^\natural$ -convex as $w_{T+1}(x_{T+1}, B_{T+1})$ is additively separable in (x_{T+1}, B_{T+1}) . For $t = 1, \dots, T$, suppose $w_{t+1}(x_{t+1}, B_{t+1})$ is $L^\natural$ -convex, we

shall show $w_t(x_t, B_t)$ is also $L^\natural$ -convex. Notice that (i) $\beta \mathbb{E}[w_{t+1}(x_t - \zeta_t - D_t, B_t - \zeta_t)]$ is submodular in (ζ_t, x_t, B_t) by the induction assumption and that submodularity preserves under expectation; (ii) $G_t(x_t - \zeta_t)$ is submodular in (x_t, ζ_t) ; and that (iii) $-c\zeta_t + K\delta(-\zeta_t)$ only depends on ζ_t . Then it follows that $G_t(x_t - \zeta_t) - c\zeta_t + K\delta(-\zeta_t) + \beta \mathbb{E}[w_{t+1}(x_t - \zeta_t - D_t, B_t - \zeta_t)]$ is submodular in (ζ_t, x_t, B_t) , thus is $L^\natural$ -convex. Since $0 \leq -\zeta_t \leq -B_t$ is a lattice, it follows from Lemma 2 of [25] that $w_t(x_t, B_t)$ is also $L^\natural$ -convex. This completes the proof. $\square$

Note that $L^\natural$ -convexity has been widely used in inventory literature to derive structures of optimal policies (see [5,25] and references therein for more details on $L^\natural$ -convexity). Given the $L^\natural$ -convexity of $v_t(x_t, z_t)$, property (2) of Lemma 6 shows that the optimal ordering quantity $q_t^*(x_t, C_t)$ in each period is not decreasing in C_t . Property (3) further indicates that the sensitivity of $q_t^*(x_t, C_t)$ with respect to C_t is bounded by one, i.e., $\frac{q_t^*(x_t, C_t + \xi) - q_t^*(x_t, C_t)}{\xi} < 1$. It implies that the optimal ordering quantity does not change too much as the remaining capacity increases. Therefore, as C_t continues to increase, the optimal ordering quantity q_t in period must be an interior point of $[0, C_t]$.

One may wonder what if C_t increases so quickly so that there are leftovers of capacity, or technically $q_t^*(x_t, C_t) < C_t$ holds for all t . For instance, the buyer commits a sufficiently large commitment prior to the selling season to cover the highest possible demand realization. The following theorem provides the answer to this question.

Theorem 7. If $q_t^*(x_t, C_t) < C_t$ for all $t = 1, \dots, T$, i.e., the buyer's committed capacity has never been used up over the planning horizon, then in each period t , there exist s_t^* and S_t^* with $s_t^* \leq S_t^*$, which are independent of C_t , the stationary (s_t^*, S_t^*) policy is optimal. In addition, the unit price of capacity in period t is $\alpha^{T-t+1} \tau_c$.

Proof. Let $q_t^* = q_t^*(x_t, C_t)$ denote the optimal ordering quantity in period t . Per assumption that $q_t^* < C_t$ for all $t = 1, \dots, T$, then q_t^* is an interior point solution that minimizes the value function $v_t(x_t, C_t)$ for all $t = 1, \dots, T$. Define $v_t^{(2)}(x_t, C_t) = \frac{\partial v_t(x_t, C_t)}{\partial C_t}$, i.e., the partial derivative of $v_t(x_t, C_t)$ with respect to variable C_t . For any given period t and the optimal ordering quantity $(q_t^*, q_{t+1}^*, \dots, q_T^*)$, we can apply the well-known envelope theorem to Eq. (8) recursively from t to T (see [14] for more details), which yields the following equations: for $t = T + 1$,

$$v_{T+1}^{(2)}(x_{T+1}, C_{T+1}) = -\tau_c, \forall x_{T+1}$$

and for $t = 1, \dots, T$,

$$\begin{aligned} v_t^{(2)}(x_t, C_t) &= \frac{\partial}{\partial C_t} [\alpha \mathbb{E}_{D_t} [v_{t+1}(x_t + q_t^* - D_t, C_t - q_t^*)]] \\ &= \alpha \mathbb{E}_{D_t} [v_{t+1}^{(2)}(x_t + q_t^* - D_t, C_t - q_t^*)] \\ &= \dots \\ &= \alpha^{T-1+t} \mathbb{E}_{D_t, \dots, D_T} \\ &\quad \times [v_{T+1}^{(2)}(x_t + \sum_{i=t}^T q_i^* - \sum_{i=1}^T D_i, C_t - \sum_{i=t}^T q_i^*)] \\ &= -\alpha^{T-1+t} \tau_c \end{aligned}$$

where the first equality is due to the envelop theorem and the second equality is due to the bounded converge theorem.

It implies that $v_t(x_t, C_t)$ must be separable in variables x_t and C_t for $t = 1, \dots, T$, hence we can write

$$v_t(x_t, C_t) = g_t(x_t) - \alpha^{T-t+1} \tau_c C_t,$$

where $g_t(x_t) = \min_{q_t \geq 0} \{G_t(x_t + q_t) + (c + \alpha^{T-t+1} \tau_c) q_t + K\delta(q_t) + \alpha \mathbb{E}[g_{t+1}(x_t + q_t - D_t)]\}$, with $g_{T+1}(x_{T+1}) = -\tau_l [x_{T+1}]^+ + \tau_b$

$[-x_{T+1}]^+$. Notice that $g_t(x_t)$ represents a classical single unit multi-period inventory system with fixed costs as in [16]. It follows that the stationary (s, S) policy is optimal. $\square$

Theorem 7 is intuitive in the sense that, if the buyer can ignore the capacity constraints when making ordering decisions, the buyer's problem in (8) is reduced to the classical single-product multi-period inventory problem with fixed ordering costs, thus the stationary (s, S) policy is optimal.

5. Bounds on parameters of the optimal policy

In the section, we characterize lower and upper bounds on parameters of the state-dependent (s, S) policy, i.e., $S_t(z_t)$ and $s_t(z_t)$ in (5) and (6), respectively. These bounds are useful to compute the optimal policy efficiently [see e.g., 20,21].

For notional simplicity, we define:

$$H_t(y_t) = G_t(y_t) + (1 - \alpha)cy_t + \alpha c\mu_t,$$

denoting the expected one-period cost with post-order inventory position y_t excluding the fixed ordering costs in period t . Let

$$\eta_t = \arg \min_{y_t} H_t(y_t)$$

denote the global minimizer of $H_t(y_t)$. In addition, we impose the following assumption on η_t .

Assumption 1. For $t = 1, \dots, T-1$, $\eta_t \leq \eta_{t+1}$.

Note that **Assumption 1** is a reduced form of Assumption (vii) of [20]. If demands, holding and backordering costs are stationary, this assumption is naturally satisfied.

The following lemma presents several properties of some related functions, i.e., $H_t(y_t)$, $f_t(x_t, z_t)$ in (4), and $U_t(x_t, z_t)$ in (7), which are useful to derive the bounds on the parameters of the optimal policy.

Lemma 8. For $t = 1, \dots, T$,

- (1) $f_t(x_t, z_t)$ is nonincreasing in x_t for any $x_t \leq \min\{\eta_t, z_t\}$;
- (2) $f_t(x_t, z_t) \leq f_t(x'_t, z_t) + K$ for any $x_t \leq x'_t \leq z_t$;
- (3) $U_t(y'_t, z_t) - U_t(y_t, z_t) \leq H_t(y'_t) - H_t(y_t) \leq 0$ for any $y_t \leq y'_t \leq \min\{\eta_t, z_t\}$;
- (4) $U_t(y'_t, z_t) - U_t(y_t, z_t) \geq H_t(y'_t) - H_t(y_t) - \alpha K$ for any $y_t \leq y'_t \leq z_t$.

Proof. We show the results by induction. For part (1), it is clear that the argument is true for $t = T$. Now suppose that part (1) is true for period $t+1$, we next show it also holds for period t . First, by inductive assumption, as $f_{t+1}(y_t - D_t, z_t - D_t)$ is nonincreasing in y_t for any realization of D_t such that $y_t - D_t \leq \min\{\eta_{t+1}, z_t - D_t\}$, then we have $\alpha \mathbb{E}[f_{t+1}(y_t - D_t, z_t - D_t)]$ is nonincreasing in y_t as monotonicity is preserved under expectation. Since $\eta_t \leq \eta_{t+1}$ and $H_t(y_t)$ is decreasing in y_t for $y_t \leq \eta_t$, it follows that $U_t(y_t, z_t) = H_t(y_t) + \alpha \mathbb{E}[f_{t+1}(y_t - D_t, z_t - D_t)]$ is also nonincreasing in y_t for $y_t \leq \min\{\eta_t, z_t\}$.

Second, define $a \vee b = \max\{a, b\}$ for $a, b \in \mathbb{R}$. Denote $y_t(x_t, z_t)$ as the optimal order-up-to level under state (x_t, z_t) , then for $x_t < x'_t \leq \min\{\eta_t, z_t\}$, we have

$$\begin{aligned} f_t(x_t, z_t) &= U_t(y_t(x_t, z_t), z_t) + K\delta(y_t(x_t, z_t) - x_t) \\ &\geq U_t(y_t(x_t, z_t) \vee x'_t, z_t) + K\delta(y_t(x_t, z_t) \vee x'_t - x'_t) \\ &\geq \min_{x'_t \leq y_t \leq z_t} \{U_t(x'_t, z_t) + K\delta(y_t - x'_t)\} \\ &= f_t(x'_t, z_t), \end{aligned}$$

where the first inequality follows from (i) that $U_t(y_t, z_t)$ is nonincreasing in y_t for $y_t \leq \min\{\eta_t, z_t\}$, and (ii) that $K\delta(y_t(x_t, z_t) - x_t) \geq K\delta(y_t(x_t, z_t) \vee x'_t - x'_t)$ for any $x'_t > x_t$; and the second inequality

follows from that $z_t \geq y_t(x_t, z_t) \vee x'_t \geq x'_t$. This completes the induction for part (1).

For part (2), one can verify that it holds for period T . Now suppose that it is true for period $t+1$. Then, for $x_t \leq x'_t$, we have

$$\begin{aligned} f_t(x_t, z_t) &= \min_{x_t \leq y_t \leq z_t} \{U_t(y_t, z_t) + K\delta(y_t - x_t)\} \\ &\leq \min_{x_t \leq y_t \leq z_t} \{U_t(y_t, z_t) + K[\delta(y_t - x'_t) + \delta(x'_t - x_t)]\} \\ &\leq K + \min_{x'_t \leq y_t \leq z_t} \{U_t(y_t, z_t) + K\delta(y_t - x'_t)\} \\ &= K + f_t(x'_t, z_t). \end{aligned}$$

where the first inequality is due to the subadditivity of $\delta(\cdot)$. Hence, part (2) also holds for period t , which completes the induction.

For part (3), note that for $y_t \leq y'_t \leq \min\{\eta_t, z_t\}$,

$$\begin{aligned} &U_t(y'_t, z_t) - U_t(y_t, z_t) \\ &= H_t(y'_t) - H_t(y_t) + \alpha \mathbb{E}[f_{t+1}(y'_t - D_t, z_t - D_t) - f_{t+1}(y_t - D_t, z_t - D_t)] \\ &\leq H_t(y'_t) - H_t(y_t) \leq 0. \end{aligned}$$

where the first inequality follows from part (1) and $\eta_t \leq \eta_{t+1}$.

For part (4), note that for $y_t \leq y'_t \leq z_t$,

$$\begin{aligned} &U_t(y'_t, z_t) - U_t(y_t, z_t) \\ &= H_t(y'_t) - H_t(y_t) + \alpha \mathbb{E}[f_{t+1}(y'_t - D_t, z_t - D_t) - f_{t+1}(y_t - D_t, z_t - D_t)] \\ &\geq H_t(y'_t) - H_t(y_t) - \alpha K, \end{aligned}$$

where the inequality is due to part (2). $\square$

We further define parameters $\bar{S}_t(z_t)$, $\underline{s}_t(z_t)$ and $\bar{s}_t(z_t)$ with $\bar{S}_t(z_t) \geq \min\{\eta_t, z_t\} \geq \min\{\underline{s}_t(z_t), \bar{s}_t(z_t)\}$ such that

$$H_t(\bar{S}_t(z_t)) = H_t(\min\{\eta_t, z_t\}) + \alpha K, \quad (9)$$

$$H_t(\underline{s}_t(z_t)) = H_t(\min\{\eta_t, z_t\}) + K, \quad (10)$$

$$H_t(\bar{s}_t(z_t)) = H_t(\min\{\eta_t, z_t\}) + (1 - \alpha)K. \quad (11)$$

Then the bounds on parameters $s_t(z_t)$ and $S_t(z_t)$ of the state-dependent (s, S) policy are summarized in the following theorem.

Theorem 9. For $t = 1, \dots, T$,

- (1) $\min\{\eta_t, z_t\} \leq S_t(z_t) \leq \bar{S}_t(z_t)$, and $S_t(z_t) = z_t$ whenever $\eta_t \geq z_t$;
- (2) $\underline{s}_t(z_t) \leq s_t(z_t) \leq \bar{s}_t(z_t)$.

Proof. For (1), recall that $S_t(z_t) = \min \arg \min_{-\infty < y_t \leq z_t} U_t(y_t, z_t)$ defined in (5), and $U_t(y_t, z_t)$ is nonincreasing in y_t on $(-\infty, \min\{\eta_t, z_t\}]$, which is shown in the proof of **Lemma 8**. Therefore, we must have $S_t(z_t) \geq \min\{\eta_t, z_t\}$. Moreover, if $z_t > \eta_t$, $\min\{\eta_t, z_t\} = \eta_t$, by part (2) of **Lemma 8**, for $y_t \geq \bar{S}_t(z_t) \geq \eta_t$, one can readily show that $U_t(y_t, z_t) - U_t(\eta_t, z_t) \geq H_t(y_t) - H_t(\eta_t) - \alpha K \geq H_t(\bar{S}_t(z_t)) - H_t(\eta_t) - \alpha K = 0$. Hence, for this case we must have $S_t(z_t) \leq \bar{S}_t(z_t)$. If $z_t \leq \eta_t$, then $S_t(z_t) = z_t \leq \bar{S}_t(z_t)$. Therefore, $S_t(z_t) \leq \bar{S}_t(z_t)$ also holds.

As for (2), note that $U_t(S_t(z_t), z_t) + K = U_t(s_t(z_t), z_t)$. To show that $s_t(z_t) \geq \underline{s}_t(z_t)$, we consider two scenarios: (i) if $S_t(z_t) \geq \eta_t$, we have

$$\begin{aligned} U_t(S_t(z_t), z_t) + K - U_t(\underline{s}_t(z_t), z_t) &\leq U_t(\eta_t, z_t) + K - U_t(\underline{s}_t(z_t), z_t) \\ &\leq H_t(\eta_t) + K - H_t(\underline{s}_t(z_t)) = 0, \end{aligned}$$

where the first inequality is due to the definition of $S_t(z_t)$ and the second inequality is due to part (3) of **Lemma 8**, and (ii) if $S_t(z_t) < \eta_t$, we must have $S_t(z_t) = z_t$ and hence

$$\begin{aligned} U_t(S_t(z_t), z_t) + K - U_t(\underline{s}_t(z_t), z_t) &= U_t(z_t, z_t) + K - U_t(\underline{s}_t(z_t), z_t) \\ &\leq H_t(z_t) + K - H_t(\underline{s}_t(z_t)) = 0, \end{aligned}$$

Table 1
 $(\underline{S}_3(z_3), \bar{S}_3(z_3), \underline{S}_3(z_3), \bar{S}_3(z_3), \underline{S}_3(z_3), \bar{S}_3(z_3))$ as a function of z_3 and b .

z_3	$b = 0.6$						$b = 0.8$						$b = 1$					
50	(7	13	21	−3	0	6)	(7	14	21	−1	1	7)	(7	14	21	1	2	7)
49	(−	−	−	−	−	−)	(−	−	−	−	−	−)	(−	−	−	−	−	−)
48	(−	−	−	−	−	−)	(−	13	−	−	−	−)	(−	−	−	−	−	−)
47	(−	12	−	−	−	−)	(−	−	−	−	2	−)	(−	−	−	−	−	−)
46	(−	−	−	−	−	−)	(−	−	−	−	−	−)	(−	13	−	−	−	−)
45	(−	−	−	−	1	−)	(−	−	−	−	−	−)	(−	−	−	−	3	−)
44	(−	−	−	−	−	−)	(−	−	−	−	−	−)	(−	−	−	−	−	−)
43	(−	−	−	−	0	−)	(−	12	−	−	−	−)	(−	−	−	−	−	−)
42	(−	11	−	−	−	−)	(−	−	−	−	−	−)	(−	−	−	−	−	−)
41	(−	−	−	−	−	−)	(−	−	−	−	−	−)	(−	−	−	−	2	−)

− suggests that the result is the same as the row immediately above.

Table 2
 $(\underline{S}_7(z_7), \bar{S}_7(z_7), \underline{S}_7(z_7), \bar{S}_7(z_7), \underline{S}_7(z_7), \bar{S}_7(z_7))$ as a function of z_7 and b .

z_7	$b = 0.6$						$b = 0.8$						$b = 1$					
10	(7	10	21	−2	5	7)	(8	10	22	0	6	8)	(8	10	22	1	7	8)
9	(−	9	−	−	6	−)	(−	9	−	−	7	−)	(−	9	−	−	−	−)
8	(−	8	−	−	−	−)	(−	8	−	−	−	−)	(−	8	−	−	−	−)
7	(−	7	−	−	5	−)	(7	7	−	−	6	7)	(7	7	−	−	6	7)
6	(6	6	22	−3	−	6)	(6	6	−	−	5	6)	(6	6	23	−	5	6)
5	(5	5	23	−	4	5)	(5	5	24	−1	4	5)	(5	5	25	0	4	5)
4	(4	4	24	−4	3	4)	(4	4	25	−2	3	4)	(4	4	27	−1	3	4)
3	(3	3	25	−5	2	3)	(3	3	27	−3	2	3)	(3	3	29	−2	2	3)
2	(2	2	27	−6	1	2)	(2	2	29	−4	1	2)	(2	2	32	−3	1	2)
1	(1	1	28	−7	0	1)	(1	1	31	−5	0	1)	(1	1	34	−4	0	1)

− suggests that the result is the same as the row immediately above.

$\eta_7 = 7, 8, 8$, when $b = 0.6, 0.8, 1$ respectively.

where the inequality is due to part (3) of Lemma 8. Therefore, we must have $s_t(z_t) \leq \bar{s}_t(z_t)$, since $U_t(y_t, z_t)$ is nonincreasing in y_t on $(-\infty, \min\{\eta_t, z_t\}]$.

To show that $s_t(z_t) \leq \bar{s}_t(z_t)$, we differentiate two scenarios: (i) if $S_t(z_t) \geq \eta_t$, we have

$$\begin{aligned} U_t(S_t(z_t), z_t) + K - U_t(\bar{s}_t(z_t), z_t) &\geq H_t(S_t(z_t)) + K - H_t(\bar{s}_t(z_t)) - \alpha K \\ &\geq H_t(\eta_t) - H_t(\bar{s}_t(z_t)) + (1 - \alpha)K \\ &= 0. \end{aligned}$$

where the first inequality is due to part (4) of Lemma 8; and (ii) if $S_t(z_t) < \eta_t$, we must have $S_t(z_t) = z_t$, and hence

$$\begin{aligned} &U_t(S_t(z_t), z_t) + K - U_t(\bar{s}_t(z_t), z_t) \\ &= U_t(z_t, z_t) + K - U_t(\bar{s}_t(z_t), z_t) \\ &\geq H_t(z_t) + K - H_t(\bar{s}_t(z_t)) - \alpha K = 0. \end{aligned}$$

Hence, we must have $s_t(z_t) \leq \bar{s}_t(z_t)$. $\square$

6. Numerical studies

In this section, we conduct extensive numerical studies to illustrate how the optimal policy and the bounds we proposed work. We also show how different parameters affect the performance of the optimal policy. These numerical studies are structured similarly as [22].

In our numerical studies, demands are normally distributed and stochastically increasing with equal variance, i.e., $D_t \sim N(\mu_t, \sigma^2)$, where $\mu_t = \mu_0 + t \times 0.2$, $\mu_0 = 5$, $\sigma = 2$. For computational simplicity, we approximate D_t using discrete distributions with fixed support $\{0, 1, 2, \dots, 10\}$: $\text{Prob}(D_t = 0) = \Phi(\frac{0-\mu_t}{\sigma})$, $\text{Prob}(D_t = i) = \Phi(\frac{i-\mu_t}{\sigma}) - \Phi(\frac{i-1-\mu_t}{\sigma})$ for $i = 1, 2, \dots, 9$, and $\text{Prob}(D_t = 10) = 1 - \Phi(\frac{9-\mu_t}{\sigma})$, and $\Phi(\cdot)$ is the cumulative distribution of the standard normal distribution. We fix $x_1 = 10$, $c = 0.5$, $h = 0.4$, $\tau_c = 0.3$, $\tau_l = 0.2$, $\tau_b = 0.6$, $\mu_0 = 5$, $\alpha = 0.99$, and vary $p \in \{0.8, 0.9, 1, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6\}$, $b \in \{0.6, 0.8, 1\}$, $K \in \{5, 10\}$, $\sigma \in \{1, 1.5, 2\}$ and $T \in \{10, 20\}$.

Note that Assumption 1 is satisfied in all settings, i.e., η_t is nondecreasing in t .

We first present our state-dependent policy $(S_t(z_t), s_t(z_t))$ and the bounds $(\underline{S}_t(z_t), \bar{S}_t(z_t), \underline{s}_t(z_t), \bar{s}_t(z_t))$, where $\underline{S}_t(z_t) = \min\{\eta_t, z_t\}$ in Theorem 9. Tables 1 and 2 show results of the optimal policy and bounds for the setting $x_0 = 10$, $K = 5$, $c = 0.5$, $h = 0.4$, $\mu_0 = 5$, $\sigma = 2$, $\tau_c = 0.3$, $\tau_l = 0.2$, $T = 10$ with $b \in \{0.6, 0.8, 1\}$. Table 1 takes period $t = 3$ as an example, where all $z_t \geq \eta_t$. Note that z_3 has 21 possible states depending on the realizations of D_1 and D_2 , e.g., $z_3 \in [33, 53]$ when $b = 0.6$. We can see that the bounds $(\underline{S}_3(z_3), \bar{S}_3(z_3), \underline{s}_3(z_3), \bar{s}_3(z_3))$ remain the same for different z_3 given b . This is because $\underline{S}_3(z_3) = \min\{\eta_3, z_3\} = \eta_3 = 7$ remains fixed and so are $\bar{S}_3(z_3), \underline{s}_3(z_3), \bar{s}_3(z_3)$ due to Eqs. (9)–(11).

In contrast, Table 2 shows the case when $z_t < \eta_t$ given $t = 7$, where $z_7 \in [0, x_1 + C_0^*]$ in our settings. We can see that the bounds $\underline{S}_7(z_7), \bar{S}_7(z_7), \underline{s}_7(z_7)$ are non-increasing as z_7 decreases from η_7 , while $\bar{S}_7(z_7)$ becomes quite loose because $H_t(y_t)$ is convex and $\bar{S}_t(z_t) > \eta_t$. The results confirm our analyses and Theorem 9. It is worth noting that the optimal policy $s_t(z_t), S_t(z_t)$ are not monotone in z_t for a given period t .

Tables 1 and 2 also show that the optimal policy and bounds are nondecreasing in the shortage cost b . We next investigate the impact of other parameters, such as K, σ , on the optimal policy and bounds given z_t similar to Tables 1 and 2. Note that we set $x_0 = 10$, $b = 0.6$, $c = 0.5$, $h = 0.4$, $\mu_0 = 5$, $\sigma = 2$, $\tau_c = 0.3$, $\tau_l = 0.2$, $T = 10$ with $K \in \{5, 10\}$ in Tables 3 and 4, and $x_0 = 10$, $K = 5$, $b = 0.6$, $c = 0.5$, $h = 0.4$, $\mu_0 = 5$, $\tau_c = 0.3$, $\tau_l = 0.2$, $T = 10$ with $\sigma \in \{2, 1.5, 1\}$ in Tables 5 and 6. Tables 3–6 show that $\underline{S}_t(z_t), \bar{S}_t(z_t)$ is non-decreasing in K, σ , while $\underline{s}_t(z_t), s_t(z_t), \bar{s}_t(z_t)$ are non-increasing in K, σ given t and z_t . However, $S_t(z_t)$ is non-monotone in σ while p does not affect the optimal policy as it only affects the optimal commitment C_0^* .

We also investigate how different parameters, such as p, K, b, σ, T , affect the buyer's optimal performance, i.e., the total expected cost V_0^* and its optimal capacity commitment C_0^* . Tables 7 and 8 show that the optimal commitment C_0^* is nondecreasing in the unit shortage cost b , but nonincreasing in the unit

Table 3 $(\underline{S}_3(z_3), S_3(z_3), \bar{S}_3(z_3), \underline{s}_3(z_3), s_3(z_3), \bar{s}_3(z_3))$ as functions of z_3 and K .

z_7	$K = 5$						$K = 10$					
50	(7	13	21	−3	0	6)	(7	19	33	−12	−4	6)
49	(−	−	−	−	−	−)	(−	−	−	−	−	−)
48	(−	−	−	−	−	−)	(−	−	−	−	−	−)
47	(−	12	−	−	−	−)	(−	18	−	−	−	−)
46	(−	−	−	−	−	−)	(−	−	−	−	−3	−)
45	(−	−	−	−	1	−)	(−	−	−	−	−	−)
44	(−	−	−	−	−	−)	(−	17	−	−	−	−)
43	(−	−	−	−	0	−)	(−	−	−	−	−	−)
42	(−	11	−	−	−	−)	(−	−	−	−	−2	−)
41	(−	−	−	−	−	−)	(−	16	−	−	−	−)

− suggests that the result is the same as the row immediately above.

Table 4 $(\underline{S}_7(z_7), S_7(z_7), \bar{S}_7(z_7), \underline{s}_7(z_7), s_7(z_7), \bar{s}_7(z_7))$ as functions of z_7 and K .

z_7	$K = 5$						$K = 10$					
10	(7	10	21	−2	5	7)	(7	10	34	−11	5	7)
9	(−	9	−	−	6	−)	(−	9	−	−	−	−)
8	(−	8	−	−	−	−)	(−	8	−	−	4	−)
7	(−	7	−	−	5	−)	(−	7	−	−	−	−)
6	(6	6	22	−3	−	6)	(6	6	−	−	3	6)
5	(5	5	23	−	4	5)	(5	5	35	−12	2	5)
4	(4	4	24	−4	3	4)	(4	4	36	−	1	4)
3	(3	3	25	−5	2	3)	(3	3	37	−13	0	3)
2	(2	2	27	−6	1	2)	(2	2	39	−14	−1	2)
1	(1	1	28	−7	0	1)	(1	1	40	−15	−2	1)

− suggests that the result is the same as the row immediately above.

Table 5 $(\underline{S}_3(z_3), S_3(z_3), \bar{S}_3(z_3), \underline{s}_3(z_3), s_3(z_3), \bar{s}_3(z_3))$ as functions of z_3 and σ .

z_3	$\sigma = 2$						$\sigma = 1.5$						$\sigma = 1$					
50	(7	13	21	−3	0	6)	(6	13	20	−3	0	6)	(6	13	20	−2	0	6)
49	(−	−	−	−	−	−)	(−	−	−	−	−	−)	(−	−	−	−	−	−)
48	(−	−	−	−	−	−)	(−	−	−	−	−	−)	(−	−	−	−	−	−)
47	(−	12	−	−	−	−)	(−	−	−	−	−	−)	(−	−	−	−	−	−)
46	(−	−	−	−	−	−)	(−	12	−	−	−	−)	(−	−	−	−	−	−)
45	(−	−	−	−	1	−)	(−	−	−	−	1	−)	(−	12	−	−	1	−)
44	(−	−	−	−	−	−)	(−	−	−	−	−	−)	(−	−	−	−	−	−)
43	(−	−	−	−	0	−)	(−	−	−	−	−	−)	(−	−	−	−	−	−)
42	(−	11	−	−	−	−)	(−	−	−	−	0	−)	(−	−	−	−	0	−)
41	(−	−	−	−	−	−)	(−	11	−	−	−	−)	(−	−	−	−	−	−)

− suggests that the result is the same as the row immediately above.

 $S_7(z_7) = 7, 8, 9$ when $\sigma = 1, 1.5, 2$, respectively, given $z_7 = 21$, hence, $S_t(z_t)$ is non-monotone in z_t .**Table 6** $(\underline{S}_7(z_7), S_7(z_7), \bar{S}_7(z_7), \underline{s}_7(z_7), s_7(z_7), \bar{s}_7(z_7))$ as functions of z_7 and σ .

z_7	$\sigma = 2$						$\sigma = 1.5$						$\sigma = 1$					
10	(7	10	21	−2	5	7)	(7	10	21	−2	5	7)	(7	10	21	−2	5	7)
9	(−	9	−	−	6	−)	(−	9	−	−	6	−)	(−	9	−	−	6	−)
8	(−	8	−	−	−	−)	(−	8	−	−	−	−)	(−	8	−	−	−	−)
7	(−	7	−	−	5	−)	(−	7	−	−	5	−)	(−	7	−	−	−	−)
6	(6	6	22	−3	−	6)	(6	6	22	−	−	6)	(6	6	−	−	5	6)
5	(5	5	23	−	4	5)	(5	5	23	−3	4	5)	(5	5	22	−3	4	5)
4	(4	4	24	−4	3	4)	(4	4	24	−4	3	4)	(4	4	24	−4	3	4)
3	(3	3	25	−5	2	3)	(3	3	25	−5	2	3)	(3	3	25	−5	2	3)
2	(2	2	27	−6	1	2)	(2	2	27	−6	1	2)	(2	2	27	−6	1	2)
1	(1	1	28	−7	0	1)	(1	1	28	−7	0	1)	(1	1	28	−7	0	1)

− suggests that the result is the same as the row immediately above.

capacity price p , the fixed cost K , and the standard deviation σ of demand distributions. It is also straightforward to see that the total expected cost V_0^* is increasing in all the parameters, i.e., b, p, K, T, σ .

7. Conclusions and discussions

We conclude by discussing possible extensions of our results. First, it is straightforward to extend our results to systems with

fixed lead times by transforming state variables. Second, our analysis can be readily carried out into systems with time-varying fixed ordering costs as long as $K_t \geq \alpha K_{t+1}$ for $t = 1, \dots, T - 1$. Third, our results can also extend to systems with stochastic price-dependent demands. For instance, by a similar argument in this paper, one can show that the state-dependent (s, S, p) policy is optimal with additive price-dependent demand using the notion of K -concavity.

Table 7
(C_0^*, V_0^*) as a function of T, b, p given $K = 5, x_1 = 10$.

(C_0^*, V_0^*)	$b = 0.6$		$b = 0.8$		$b = 1$	
	$T = 10$	$T = 20$	$T = 10$	$T = 20$	$T = 10$	$T = 20$
p						
0.8	(43 100.34)	(122 240.43)	(46 104.82)	(127 247.63)	(48 108.18)	(130 252.50)
0.9	(41 104.54)	(121 252.58)	(45 109.41)	(126 260.28)	(48 112.98)	(129 265.48)
1.0	(39 108.52)	(119 264.54)	(45 113.91)	(124 272.81)	(47 117.69)	(128 278.35)
1.1	(37 112.31)	(117 276.33)	(44 118.32)	(123 285.18)	(46 122.34)	(127 291.12)
1.2	(36 116.00)	(115 287.95)	(43 122.63)	(122 297.42)	(45 126.94)	(126 303.77)
1.3	(35 119.58)	(113 299.39)	(42 126.86)	(120 309.52)	(45 131.44)	(125 316.32)
1.4	(34 123.05)	(112 310.63)	(41 130.98)	(119 321.49)	(44 135.87)	(124 328.76)
1.5	(33 126.41)	(110 321.70)	(39 135.00)	(118 333.33)	(43 140.25)	(123 341.09)
1.6	(32 129.68)	(108 332.58)	(38 138.85)	(117 345.05)	(43 144.55)	(122 353.32)

Table 8
(C_0^*, V_0^*) as a function of p, σ, K given $x_1 = 10, b = 0.6$.

(C_0^*, V_0^*)	$\sigma = 1$		$\sigma = 1.5$		$\sigma = 2$	
	$K = 5$	$K = 10$	$K = 5$	$K = 10$	$K = 5$	$K = 10$
p						
0.8	(43 99.25)	(41 109.49)	(43 99.92)	(41 109.98)	(43 100.34)	(41 110.21)
0.9	(42 103.50)	(40 113.56)	(42 104.19)	(40 114.02)	(41 104.54)	(40 114.23)
1.0	(41 107.63)	(39 117.51)	(40 108.28)	(39 117.94)	(39 108.52)	(39 118.13)
1.1	(39 111.65)	(38 121.33)	(38 112.14)	(38 121.75)	(37 112.31)	(37 121.93)
1.2	(36 115.42)	(37 125.04)	(37 115.86)	(37 125.47)	(36 116.00)	(36 125.62)
1.3	(35 118.99)	(35 128.63)	(35 119.46)	(30 129.05)	(35 119.58)	(30 128.99)
1.4	(34 122.48)	(29 131.64)	(34 122.95)	(29 131.98)	(34 123.05)	(29 131.91)
1.5	(34 125.88)	(28 134.48)	(33 126.33)	(28 134.85)	(33 126.41)	(28 134.76)
1.6	(33 129.18)	(27 137.26)	(32 129.61)	(27 137.64)	(32 129.68)	(27 137.53)

Acknowledgments

The authors would like to thank the Area Editor, the anonymous referee for their very helpful and constructive comments which improved the paper. This work was supported by the National Natural Science Foundation of China [Grant 71572145].

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